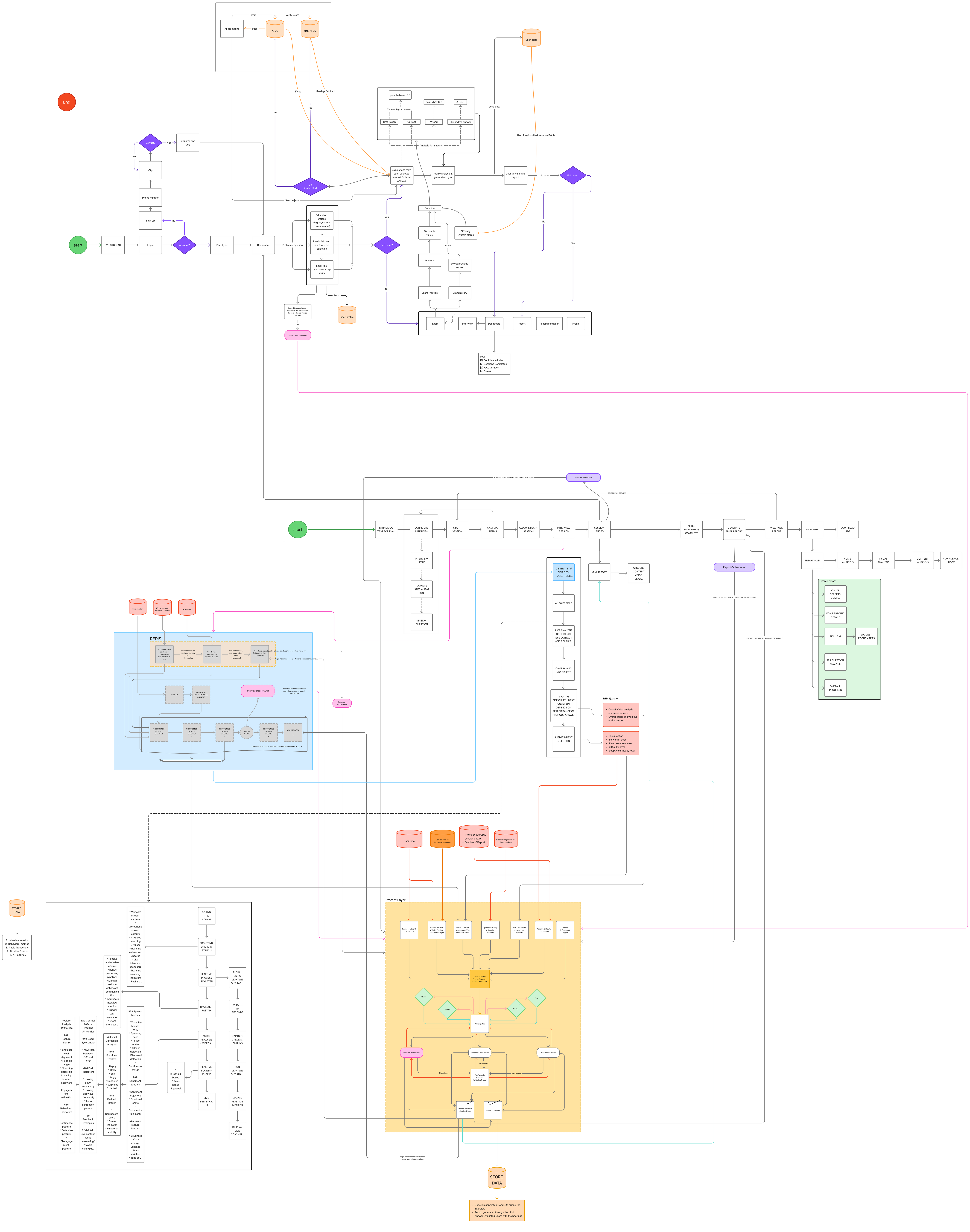
store
verify-store
AI-prompting
if No
AI QS
Non-AI QS
No
Yes
if yes
fixed qs fetched
End
point between 0-1
Time Anlaysis
points b/w 0-5
0 point
Time Taken
Correct
Wrong
Skipped/no answer
Analysis Parameters
send data
user-stats
User Previous Performance Fetch
Correct?
Yes
No
Full name and Dob
Otp
Phone number
Sign Up
No
start
B2C STUDENT
Login
account?
Plan Type
Dashboard
Profile completion
Education Details (degree/course, current marks)
1 main field and min 3 Interest selection
Email Id & Username + otp verify
Send
user-profile
Check if the questions are available in the Database on the user-selected Interest Section
no
Interview Orchestratorh
Qs Availability?
Send in json
4 questions from each selected interest for level analysis
Profile analysis & generation by AI
User gets instant report.
if old user
Full report
No
Yes
new-user?
Yes
No
Combine
Qs counts 10-30
Difficulty System stored
Re-Take
interests
select previous session
Exam Practice
Exam history
Exam
Interview
Dashboard
report
Recommendation
Profile
see:
[1] Confidence Index
[2] Sessions Completed
[3] Avg. Duration
[4] Streak
To generate basic feedback for the user/ MINI Report
Feedback Orchestrator
START NEW INTERVIEW
start
INITIAL MCQ TEST FOR EVAL
CONFIGURE INTERVIEW
INTERVIEW TYPE
DOMAIN/ SPECIALIZATION
SESSION DURATION
START SESSION
CAM/MIC PERMS
ALLOW & BEGIN SESSION
INTERVIEW SESSION
SESSION ENDED
AFTER INTERVIEW IS COMPLETE
GENERATE FINAL REPORT
VIEW FULL REPORT
OVERVIEW
DOWNLOAD PDF
BREAKDOWN
VOICE ANALYSIS
VISUAL ANALYSIS
CONTENT ANALYSIS
CONFIDENCE INDEX
Detailed report
VISUAL SPECIFIC DETAILS
VOICE SPECIFIC DETAILS
SKILL GAP
SUGGEST FOCUS AREAS
PER QUESTION ANALYSIS
OVERALL PROGRESS
Report Orchestrator
GENERATING FULL REPORT BASED ON THE INTERVIEW
PROMPT LAYER RETURNS COMPLETE REPORT
MINI REPORT
CI SCORE CONTENT VOICE VISUAL
GENERATE AI/ VERIFIED QUESTIONS...
ANSWER FIELD
LIVE ANALYSIS CONFIDENCE EYE CONTACT VOICE CLARIT...
CAMERA AND MIC OBJECT
ADAPTIVE DIFFICULTY - NEXT QUESTION DEPENDS ON PERFORMANCE OF PREVIOUS ANSWER
SUBMIT & NEXT QUESTION
REDIS(cache)
Overall Video analysis our entire session.
Overall audio analysis our entire session.
The question
answer for user
time taken to answer
difficulty level
adaptive difficulty level
Intro question
NON AI question / Validated Question
AI question
REDIS
First check in the database if questions are available Non-AI table
no question found/ total count is less than the required
Check if the questions are available in AI table
no question found/ total count is less than the required
Questions are not available in the database To conduct an interview Call the interview orchestrator
Requested number of questions to conduct an interview.
INTRO QN
FOLLOW UP COVER QN BASED ON INTRO
INTERVIEW ORCHESTRATOR
Intermediate questions based on previous answered question in interview
Interview Orchestrator
QNS FROM DB (DOMAIN SPECIFIC) 1
QNS FROM DB (DOMAIN SPECIFIC) 2
QNS FROM DB (DOMAIN SPECIFIC) 3
TRIGGER AI EVAL
QNS FROM DB (DOMAIN SPECIFIC) 4
AI GENERATED 5
In next iteration Qn 4, 5 and next Question becomes new Qn 1, 2 ,3
User data
Core persona and behavioral boundaries
Previous interview session details
Feedback/ Report
subscription profiles and feature policies
Prompt Layer
Intercept & Guard Check Trigger
Context Isolation & "Entity Tagging" (Pre-Processing)h
Stateful Context Maintenance (The Memory Tracker)
Operational Gating & Security Injections
Non-Verbal Data Structuring & Synthesis
Adaptive Difficulty Configuration
Schema Enforcement Trigger
The "Sandwich" Prompt Assembly (prompt_builder.py)
Claude
Gemini
Chatgpt
Grok
API Dispatch
Interview Orchestrator
Feedback Orchestrator
Report orchestrator
First trigger.
First trigger.
First trigger
The Pydantic Structural Validation Trigger
The Active Session Injection Trigger
The DB Committer
Requested intermediate question based on previous questions
STORE DATA
Question generated from LLM during the interview
Report generated through the LLM.
Answer Evaluated Score with the beer bag
STORED DATA
1. Interview session
2. Behavioral metrics
3. Audio Transcripts
4. Timeline Events
5. AI Reports...
BEHIND THE SCENES
FRONTEND CAM/MIC STREAM
* Webcam stream capture * Microphone stream capture * Chunked recording (5–10 sec) * Realtime websocket updates * Live interview dashboard * Realtime coaching indicators * Final ana...
REALTIME PROCESSING LAYER
FLOW - USING LIGHTWEIGHT MO...
BACKEND - FASTAPI
* Receive audio/video chunks * Run AI processing pipelines * Manage realtime websocket communication * Aggregate interview metrics * Trigger LLM evaluation * Store interview...
EVERY 5 - 10 SECONDS
AUDIO ANALYSIS + VIDEO A...
CAPTURE CAM/MIC CHUNKS
REALTIME SCORING ENGINE
RUN LIGHTWEIGHT ANA...
* Threshold-based * Rule-based * Lightwei...
LIVE FEEDBACK UI
UPDATE REALTIME METRICS
DISPLAY LIVE COACHIN...
### Speech Metrics * Words Per Minute (WPM) * Speaking pace * Pause duration * Silence detection * Filler word detection * Confidence trends ### Sentiment Metrics * Sentiment trajectory * Emotional shifts * Communication clarity ### Voice Feature Metrics * Loudness * Vocal energy variance * Pitch variation * Tone co...
## Facial Expression Analysis ### Emotions Tracked * Happy * Calm * Sad * Angry * Confused * Surprised * Neutral ### Derived Metrics * Composure score * Stress indicator * Emotional stability...
Eye Contact & Gaze Tracking ## Metrics ### Good Eye Contact * Yaw/Pitch between -10° and +10° ### Bad Indicators * Looking down repeatedly * Looking sideways frequently * Long distraction periods ## Feedback Examples * "Maintain eye contact while answering" * "Avoid looking do...
Posture Analysis ## Metrics ### Posture Signals * Shoulder level alignment * Head tilt angle * Slouching detection * Leaning forward/ backward * Engagement estimation ### Behavioral Indicators * Confidence posture * Defensive posture * Disengagement posture